



Sarvajani Education Society
Sarvajani College of Engineering & Technology, Surat.
Department of Computer Applications
Academic Year: 2026 – 27



A Project On
“BlueGuard”

Presented By
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M.C.A. II Sem. 4

Under the Guidance of
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AGENDA

- Company Profile (in brief)
- System Description (Technologies used)
- Project Overview
- Database Design/ File System (Table Designs)
- Diagrams
- Screen Shots

COMPANY PROFILE



- NJ Technologies (Finlogic Technologies Pvt. Ltd.) is a technology services company of the NJ Group, delivering secure, scalable, and business-focused IT solutions across multiple domains.
- Backed by Finlogic Technologies (est. 2002), the company over two decades of experience in managing enterprise-scale systems.
- NJ Technologies follows a solution based on simplicity, scalability, security, and success.
- The company maintains strong governance and ethical, process-driven practices to deliver reliable and trustworthy solutions.

SYSTEM DESCRIPTION

Category	Technologies
Development Framework	Flask
Database System	Mongo DB
Frontend Technology	HTML5, CSS3, JS and Jinja2
Backend Technology	Python and Wazuh
AI/ML	Gen LLM for Advance Threat Analysis
Data Storage	JSON-based NOSQL

PROJECT OVERVIEW

- BlueGuard is an blue team–oriented project designed to simulate the working of a Security Operations Center (SOC).
- The project focuses on monitoring system activities, detecting security threats, and assisting incident response using automation and analytics
- BlueGuard is built using ethical and defensive security practices and does not include any offensive or hacking-related activities.



BlueGuard

PROJECT OVERVIEW



- BlueGuard continuously collects system and security logs from endpoints using Wazuh SIEM.
- These logs are analyzed to identify suspicious activities such as failed login attempts, abnormal processes, and policy violations.
- Detected events are converted into security alerts and visualized on a centralized dashboard for analysis and monitoring.

DATABASE DESIGN

1. Raw_alert

Field Name	Data Type	Description
timestamp	DateTime	The exact time when the security event was generated.
agent.name	String	Hostname of the source machine where the alert originated.
agent.ip	String	The internal IP address of the source server.
rule.id	Integer	The unique ID assigned to the specific security rule in Wazuh.
rule.level	Integer	The internal severity level (0-15) defined by the SIEM.
rule.description	String	The generic human-readable description of the log event.
full_log	Text	The complete raw log message for deep technical investigation.

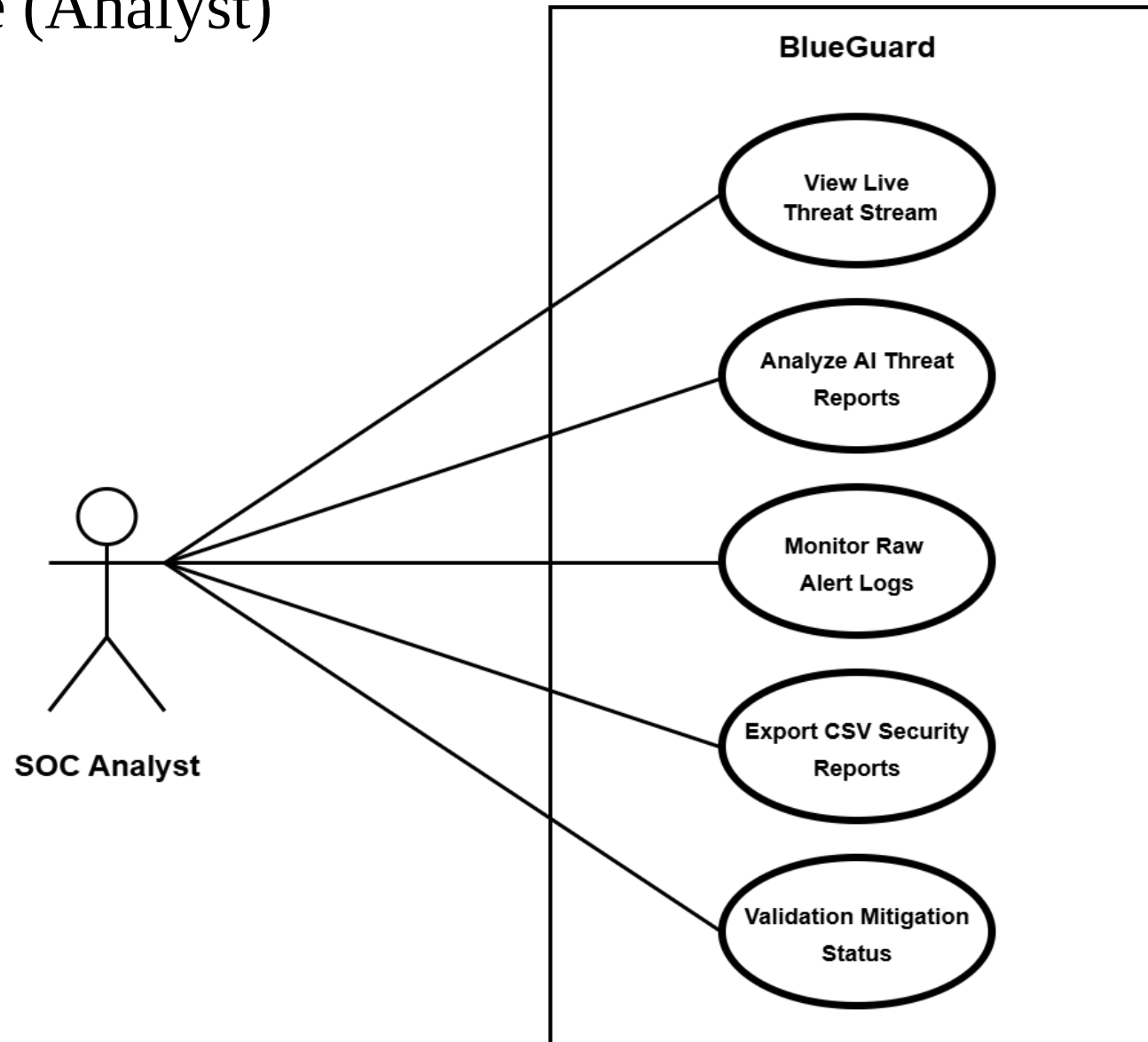
DATABASE DESIGN

2. AI_Intelligence

Field Name	Data Type	Description
attack_type	String	AI-classified category of the security incident.
mitre_tactic	String	Corresponding MITRE ATT&CK tactic (e.g., Persistence).
mitre_technique	String	Specific adversary technique and its unique ID.
base_severity	Enum	The global generic risk level of the threat.
org_risk_severity	Enum	The adjusted risk level considering internal controls.
analysis	Text	A concise 1-2 sentence AI summary of the threat.
cve_cwe	String	Hyperlinked identifiers for specific vulnerabilities.
cvss_score	Float	Technical impact score of the associated vulnerability.
remediation	Text	Step-by-step instructions generated by AI to fix the issue.

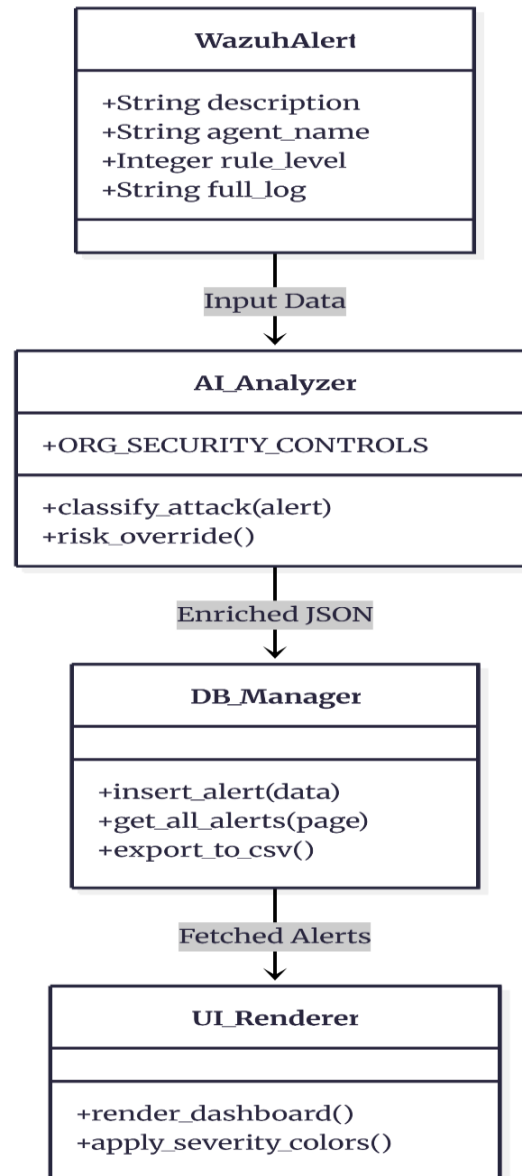
DIAGRAMS

1. Use Case (Analyst)



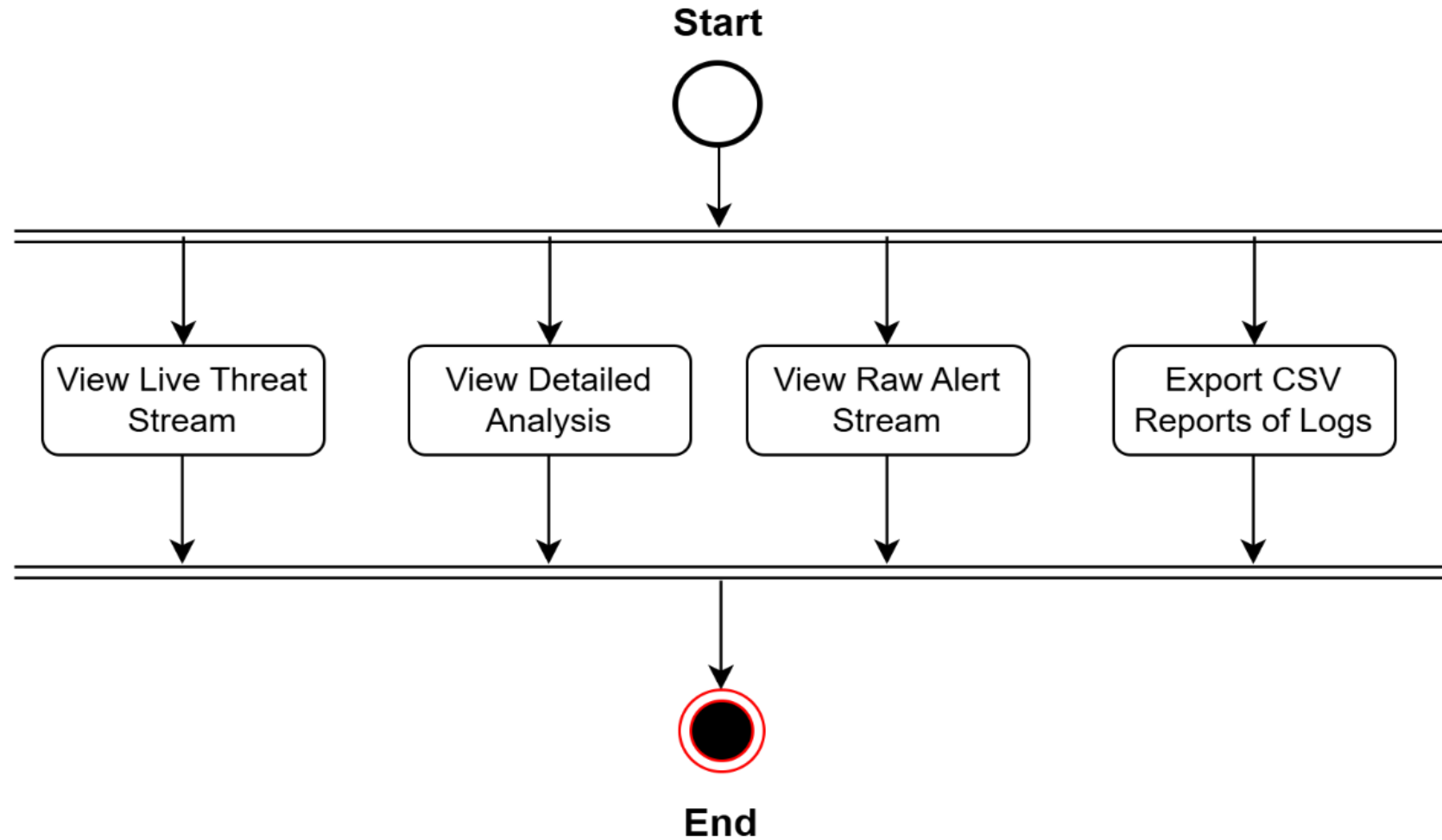
DIAGRAMS

3. Class Diagram



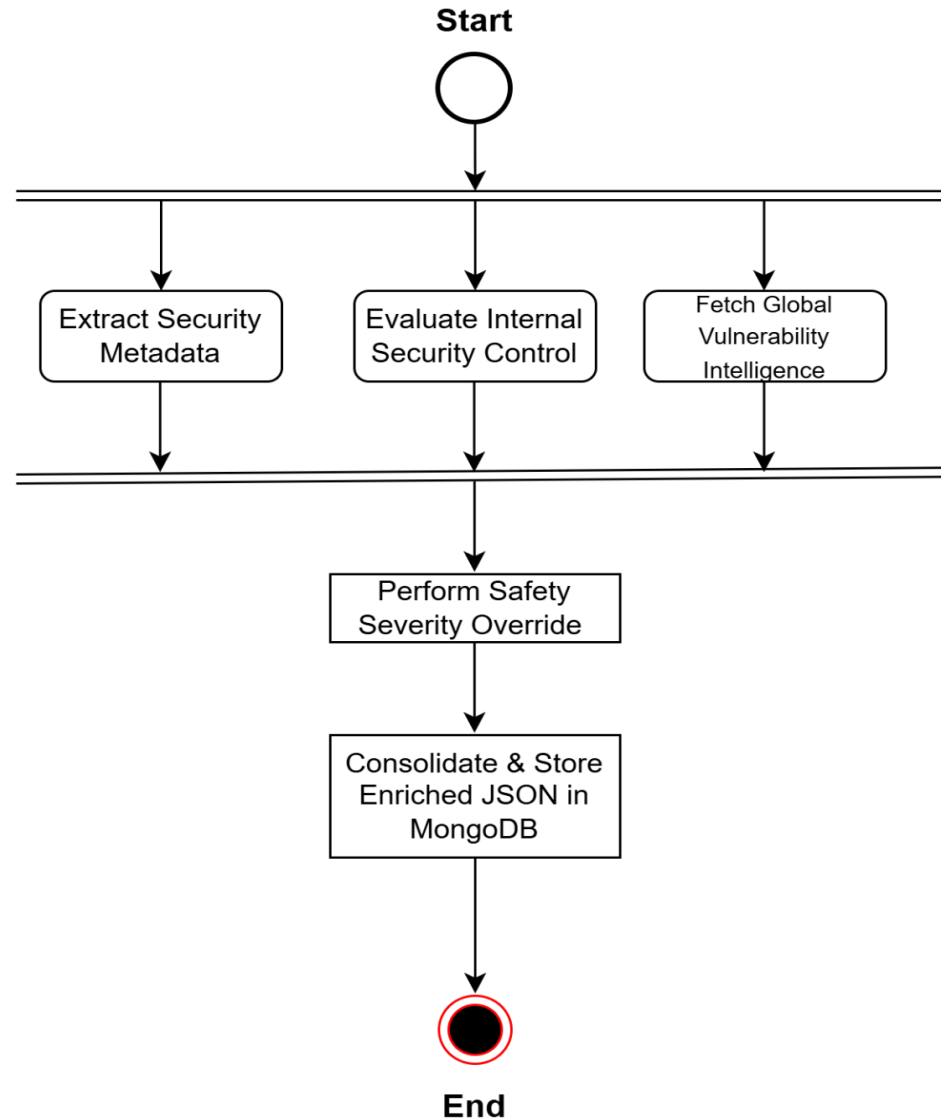
DIAGRAMS

4. Activity Diagram(App Flow)



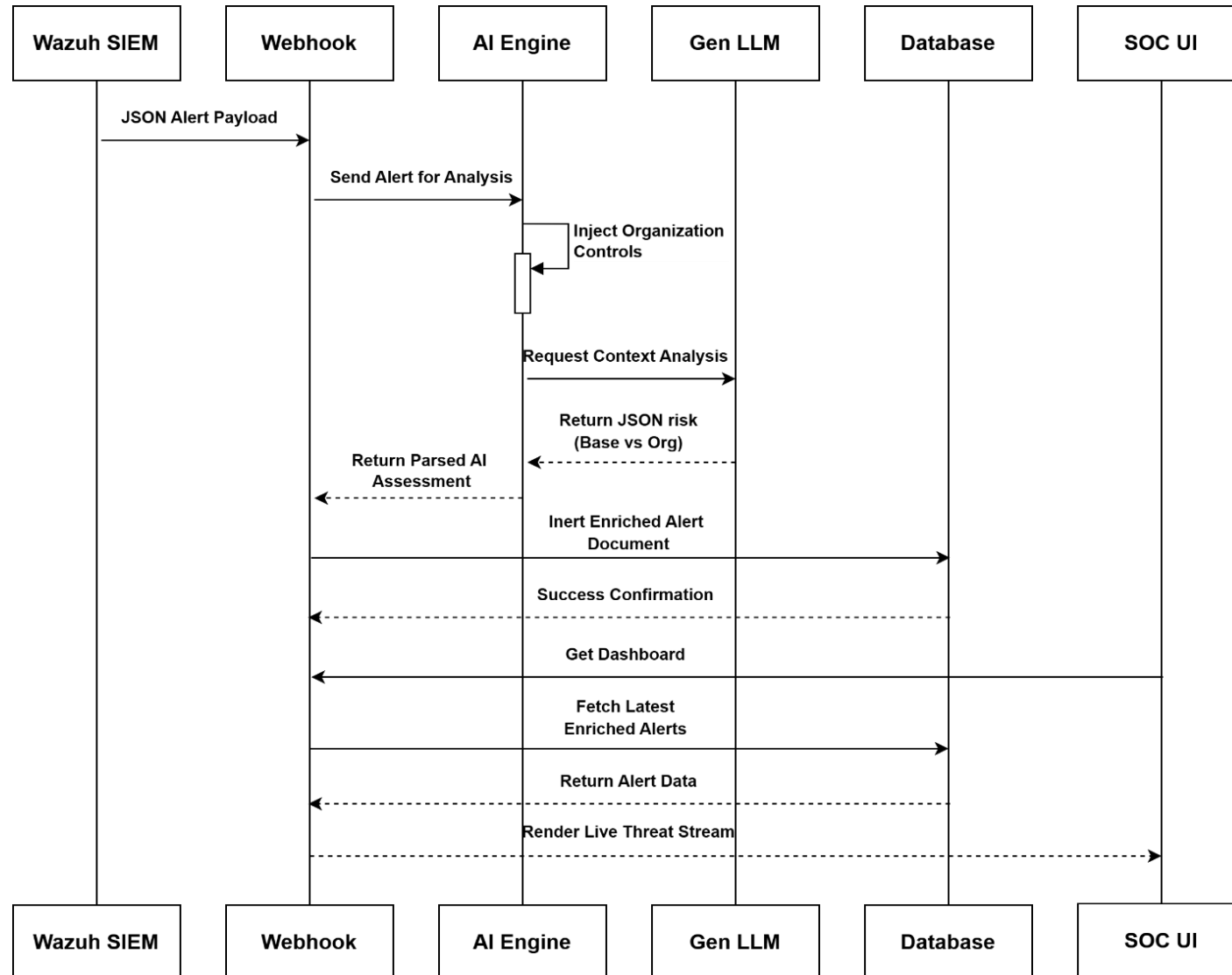
DIAGRAMS

4. Activity Diagram(AI Decision Making Logic Flow)



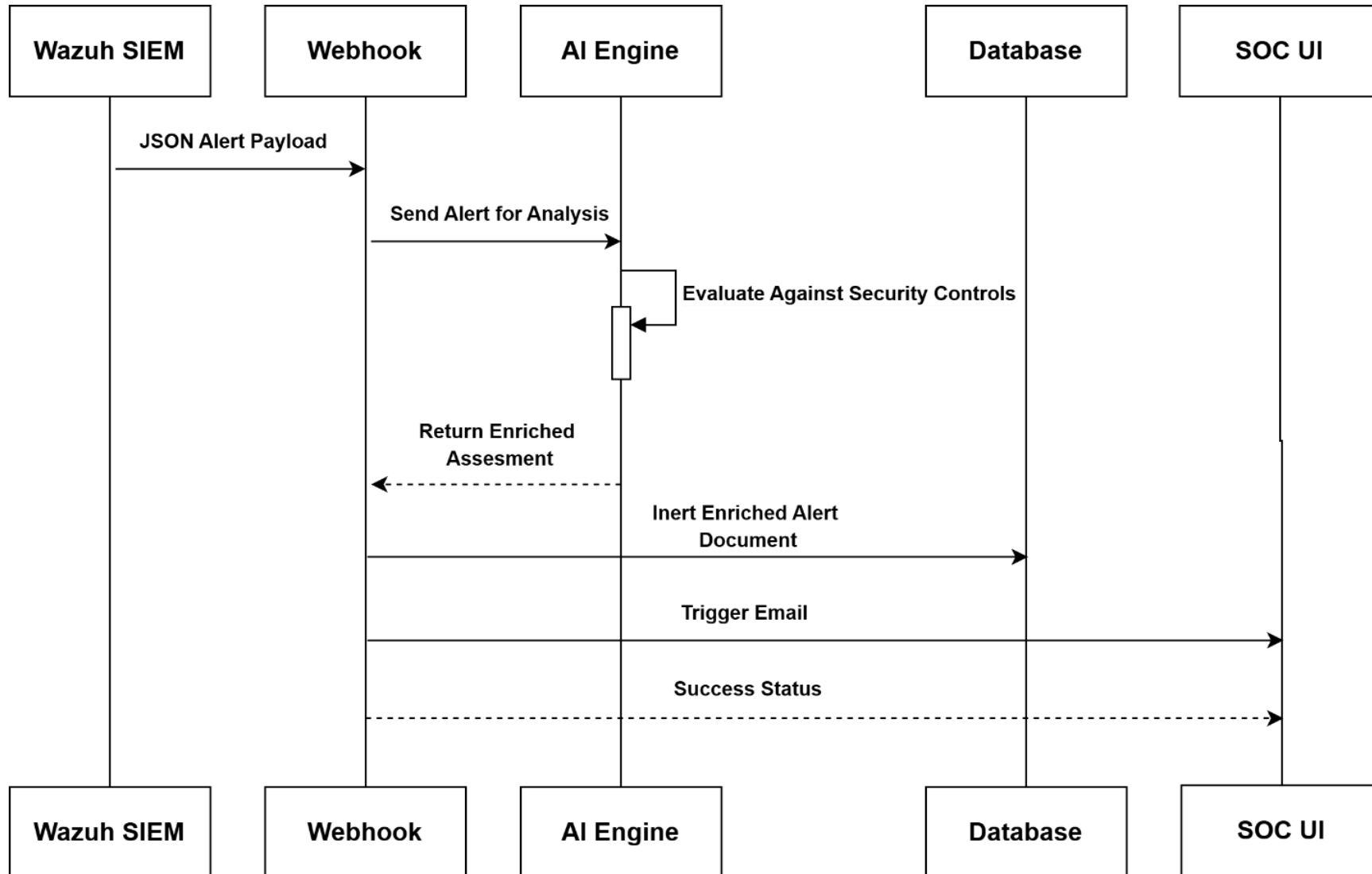
DIAGRAMS

5. Interaction Diagram (Analyst Flow)



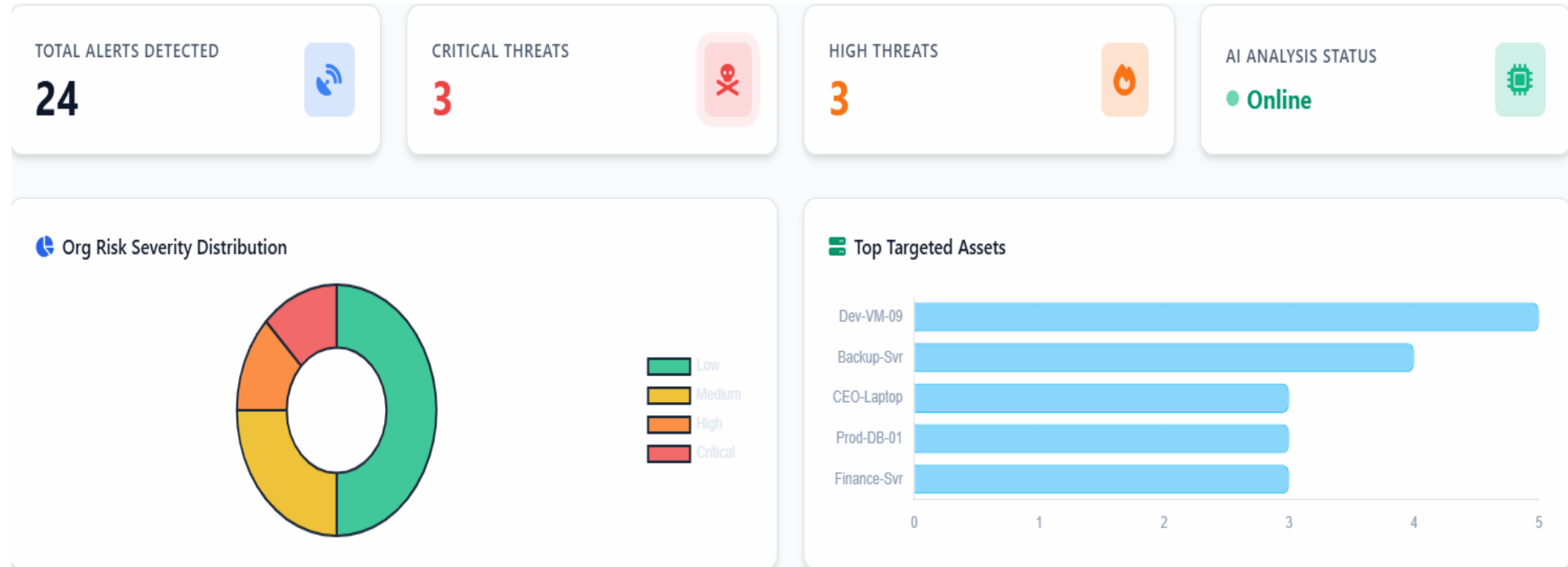
DIAGRAMS

5. Interaction Diagram (Threat Intelligence Flow)



SCREENSHOTS

1. Dashboard(Charts)



SCREENSHOTS

1. Dashboard(Alerts)

Timestamp	Agent	Source IP	AI Classification	CVE / CVSS	MitRE ATT&CK	Base Risk	Org Risk
2026-05-01 23:26:28	Backup-Svr	199.121.2.150	Brute Force Attempt HIGH: SSH Brute Force attempt on primary ...	CWE-307	T1110 CREDENTIAL ACCESS	High	Low ↓ Mitigated
2026-05-01 23:26:24	CEO-Laptop	126.101.206.51	Malware Execution CRITICAL: Malware LockBit process executi...	CWE-94 CVSS 8.0	T1203 EXECUTION	Critical	Critical
2026-05-01 23:26:21	Gateway-FW	63.246.112.83	SQL Injection HIGH: Potential SQL Injection attack detected	CWE-89 CVSS 6.5	T1190 INITIAL ACCESS	High	Medium ↓ Mitigated
2026-05-01 23:26:17	App-Server-03	218.112.85.52	Peripheral Device Connection LOW: New USB device connected	CWE-20	T1203 DEFENSE EVASION	Low	Low
2026-05-01 23:26:14	Prod-DB-01	209.42.195.186	File Modification CRITICAL: Ransomware WannaCry encrypti...	CWE- encryption CVSS 8.0	T1485 IMPACT	Critical	Critical
2026-05-01 23:26:54	Dev-VM-09	49.216.28.170	Port Scanning MEDIUM: Port scanning activity from intern...	CWE-200	T1046 RECONNAISSANCE	Medium	Low ↓ Mitigated
2026-05-01 23:26:51	Dev-VM-09	193.95.120.180	Malware Execution CRITICAL: Malware LockBit process executi...	CWE-94 CVSS 7.5	T1203 EXECUTION	Critical	Critical
2026-05-01 23:26:47	Backup-Svr	143.218.96.74	Brute Force Attempt MEDIUM: Multiple failed sudo attempts	CWE-307 CVSS 4.0	T1110 CREDENTIAL ACCESS	Medium	Low ↓ Mitigated
2026-05-01 23:26:44	App-Server-03	159.53.215.36	Legitimate Login LOW: Legitimate Administrator login	CWE-287	T1078 CREDENTIAL ACCESS	Low	High ↓ Mitigated
2026-05-01 23:26:40	HR-Portal	215.72.13.165	Brute Force Attempt MEDIUM: Multiple failed sudo attempts	CWE-307	T1110 CREDENTIAL ACCESS	Medium	Low ↓ Mitigated
Showing Page 1 of 5						Previous	Next

SCREENSHOTS

2. Detailed AI Analysis Page

 **Detailed AI Threat Analysis**

 2026-04-28 19:41:23

General Security Event

 DMZ-WEB-SERVER (10.0.1.20)

ORIGINAL WAZUH ALERT

Rule Group: web, path_traversal

Rule Level: 14

Description: Path Traversal attempt (LFI) detected.

AI ANALYSIS

 Fallback classification used because AI generation failed. Organizational controls factored into risk assessment.

VULNERABILITY INTELLIGENCE

 **CVE/CWE:** CWE-28

 **CVSS:** N/A

Base: Medium → Org: Medium

MITRE ATT&CK MAPPING

 **Tactic:** Execution

 **Technique:** User Execution (T1204)

ORGANIZATIONAL RISK ASSESSMENT

 Security event detected. While we have multiple layers of defense, the nature of this event requires manual review to ensure internal integrity is not compromised.

REMEDIATION STEPS

 Investigate manually for unusual system activity.

SCREENSHOTS

3. Live Alert Stream

LIVE RAW ALERTS

▼

All Agents

▼

mm/dd/yyyy --:-- --

mm/dd/yyyy --:-- --

Auto-refreshing 5s

[\[Back to Dashboard\]](#)

2026-04-28 19:41:23

[BASE: MEDIUM] → [ORG: MEDIUM]

Host: DMZ-WEB-SERVER | Type: General Security Event | CWE-20

CMD: Path Traversal attempt (LFI) detected.

🔒 Security event detected. While we have multiple layers of defense, the nature of this event requires manual review to en...

2026-04-28 19:38:23

[BASE: HIGH] → [ORG: MEDIUM] ↓ MITIGATED

Host: WKS-DEV-04 | Type: Ransomware Detection | CVE-2017-0144, CWE-20 | CVSS: 8.1

CMD: WannaCry ransomware indicators detected (SMB exploit).

🔒 Given the organization's robust security controls, including internal network segmentation and a strong firewall, the li...

2026-04-28 19:35:23

[BASE: HIGH] → [ORG: MEDIUM] ↓ MITIGATED

Host: DC-PRIMARY | Type: Named Pipe Creation | CWE-20

CMD: Suspicious Named Pipe created (Cobalt Strike indicator).

🔒 Given the robust security controls in place, including internal network isolation and real-time monitoring, the risk of ...

2026-04-28 19:32:23

[BASE: MEDIUM] → [ORG: MEDIUM]

Host: PROD-DB-CLUSTER | Type: Unauthorized Access | CWE-284 | CVSS: 5.0

CMD: Physical badge access during abnormal hours.

🔒 The risk is mitigated by the lack of internet exposure and strict access controls, reducing the likelihood of external e...

SCREENSHOTS

4. Exported CSV Report

Timestamp	Base Seve	Org Risk S	Agent	Attack Typ	CVE/CWE	CVSS Score	MITRE Tac	MITRE Tec	Rule Descr	AI Analysis	Org Risk A	Remediation					
2026-04-2	Medium	Low	K8S-WOR	Cross-Site	N/A	N/A	Initial Acc	Cross-Site	Cross-Site	The alert i	Given that	Monitor for similar requests and ensure web application firewalls are config					
2026-04-2	High	Medium	WKS-DEV	Credential	N/A	N/A	Credential	Credential	Mimikatz	The execu	Given the	Investigate the source of the Mimikatz execution and ensure that no unauth					
2026-04-2	High	Medium	PROD-DB	DNS Tunn	N/A	N/A	Command	Applicatio	Suspicious	The alert i	Given the	Investigate the DNS queries for malicious domains and block them at the fir					
2026-04-2	High	Medium	PROD-DB	Unauthori	N/A	N/A	Initial Acc	Valid Acco	Physical b	This event	The exploi	Review badge access logs for anomalies and conduct an employee interview					
2026-04-2	High	Medium	WKS-DEV	Data Exfilt	N/A	N/A	Exfiltratio	Exfiltratio	Mass dow	The event	The organ	Investigate the user activity on WKS-DEV-04 and implement stricter controls					
2026-04-2	High	Medium	WKS-DEV	Firewall M	N/A	N/A	Defense E	Disable Se	Windows	The alert i	Given that	Re-enable the Windows Defender Firewall and review user permissions to p					
2026-04-2	Medium	Low	DMZ-WEB	Cross-Site	CWE-79	6.1	Initial Acc	Drive-by C	Cross-Site	The alert i	Given the	Implement input validation and output encoding on web applications to mit					
2026-04-2	High	Medium	AWS-JUM	Impossible	N/A	N/A	Credential	Account M	Impossible	The alert i	Given the	Investigate the login attempts for the user 'admin' and verify the legitimacy					
2026-04-2	High	Medium	AWS-JUM	Named Pip	N/A	N/A	Persistenc	Create or	Suspicious	The creati	Given the	Investigate the source of the named pipe creation and monitor for further s					
2026-04-2	High	Low	WKS-DEV	Cross-Site	CWE-79	N/A	Initial Acc	Cross-Site	Cross-Site	The alert i	Given the	Implement input validation and sanitization on user inputs; review web app					
2026-04-2	High	Medium	WKS-DEV	File Down	N/A	N/A	Command	Applicatio	Certutil us	The event	Given the	Block the external IP address and investigate the source of the command ex					
2026-04-2	High	Medium	DMZ-WEB	Impossible	N/A	N/A	Credential	Account M	Impossible	The alert i	Given the	Investigate the login attempts for the user 'admin' and verify the legitimacy					
2026-04-2	High	Medium	AWS-JUM	Path Trave	CWE-22	N/A	Credential	Path Trave	Path Trave	The alert i	Given that	Review web application code for vulnerabilities and implement input validat					
2026-04-2	High	Medium	DC-PRIMA	Impossible	N/A	N/A	Credential	Account M	Impossible	The alert i	Given the	Investigate the login activity for the 'admin' account and enforce multi-facto					
2026-04-2	High	Medium	PROD-DB	Unauthori	N/A	N/A	Privilege E	Create Acc	AWS Clou	The alert i	Given the	Investigate the source of the request and ensure that IAM policies are corre					
2026-04-2	Critical	Medium	K8S-WOR	Ransomw	CVE-2017-	8.1	Initial Acc	Exploiticat	WannaCry	The alert i	Given the	Monitor network traffic for unusual SMB activity; ensure all systems are pat					
2026-04-2	Medium	Low	K8S-WOR	General Se	N/A	N/A	Execution	User Exec	Mass dow	Fallback d	General se	Investigate manually for unusual system activity.					
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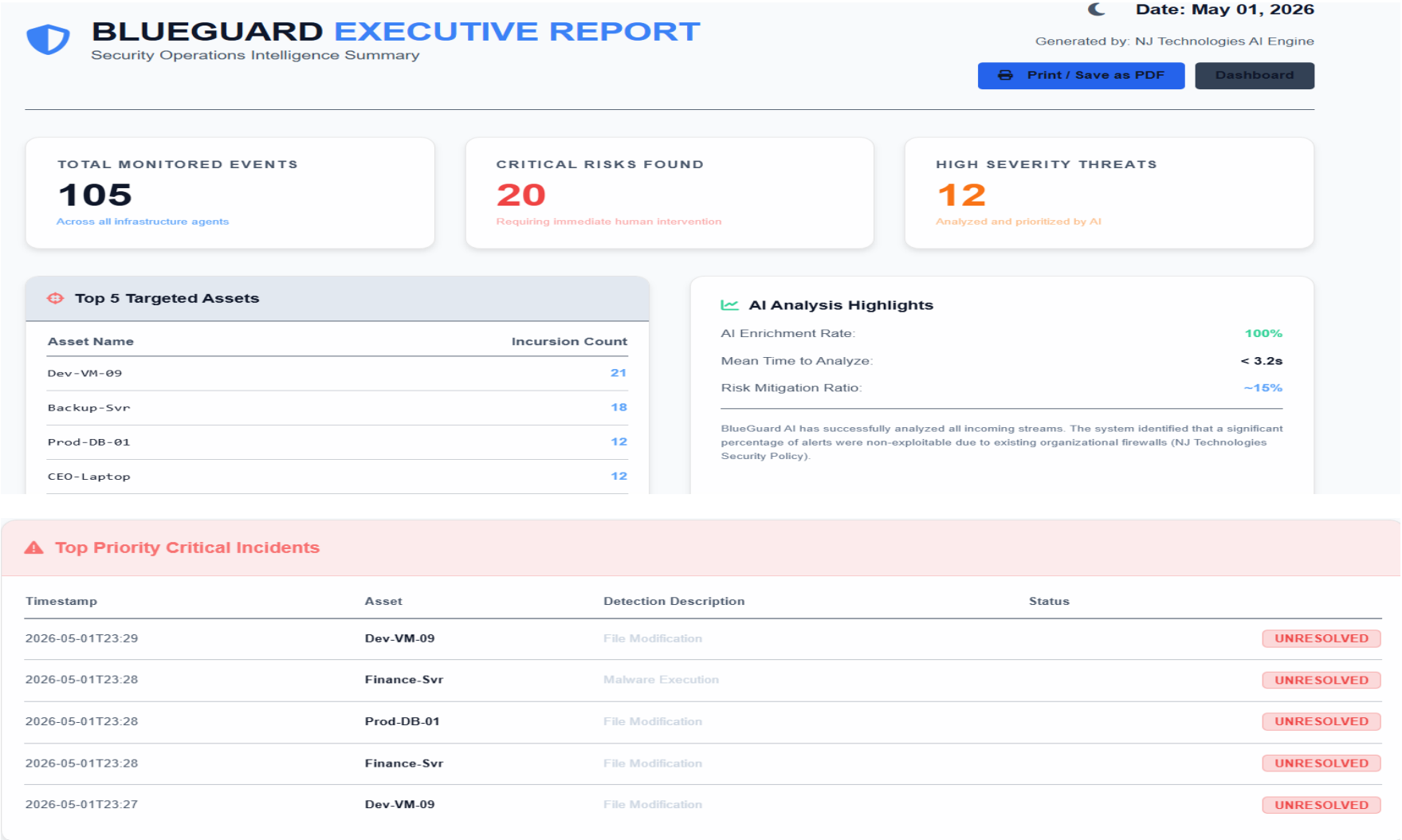
SCREENSHOTS

4. CSV Export Report

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SCREENSHOTS

5. Security Report



SCREENSHOTS

6. Asset Inventory

Infrastructure Health

Real-time risk monitoring for all discovered endpoints.



STATUS
Critical

App-Server-03

60.60.183.97

MACHINE HEALTH 0%

TOTAL ALERTS

10

CRITICAL

15

Added: N/A (Discovered)

[View Logs →](#)



STATUS
Critical

Backup-Svr

80.91.84.164

MACHINE HEALTH 0%

TOTAL ALERTS

11

CRITICAL

15

Added: N/A (Discovered)

[View Logs →](#)



STATUS
Critical

CEO-Laptop

210.68.190.103

MACHINE HEALTH 0%

TOTAL ALERTS

15

CRITICAL

15

Added: N/A (Discovered)

[View Logs →](#)

THANK YOU